DAY3 ASSIGNMENT

1.Do the following in a single program using built-in functions Take a list of numbers as input and print the largest and smallest numbers in the list. Take a string as input and print the length of the string. Take a list of names as input and print the list in alphabetical order. Take a list of numbers as input and print the total sum of the elements in the list. Takes a string as input and print the string in uppercase.

Code:

```
# Take numbers as input for largest and smallest
user_nums = input("Enter numbers separated by spaces: ")

# Convert the input string into a list of integers
nums = [int(x) for x in user_nums.split()]
print("Largest number:", max(nums))
print("Smallest number:", min(nums))

# Take string as input for length
text = input("\nEnter a string: ")
print("Length of string:", len(text))

# Take names as input for alphabetical sorting
user_names = input("\nEnter names separated by spaces: ")
names = user_names.split()
print("Sorted names:", sorted(names))

# Take numbers as input for total sum
user_sum_nums = input("\nEnter numbers to sum, separated by spaces: ")
nums_to_sum = [int(x) for x in user_sum_nums.split()]
print("Sum of list:", sum(nums_to_sum))
```

```
# Take string as input for uppercase
upper_text = input("\nEnter a string to cap: ")
print("Uppercase:", upper_text.upper())
```

Output:

```
Enter numbers separated by spaces: 5 7 8 6
Largest number: 8
Smallest number: 5
Enter a string: Amman
Length of string:
Enter names separated by spaces: Amman Farooque Sumit
Sorted names: ['Amman', 'Farooque', 'Sumit']
Enter numbers to sum, separated by spaces: 1 2 6 8 7 5
Sum of list: 29
Enter a string to cap: amman farooque
Uppercase: AMMAN FAROOQUE
```

2.Write a user-defined function factorial(n) that takes a positive integer n as an argument and returns its factorial. Use the function in a program and print the factorial of a given number.

Code:

```
# User-defined function
def factorial(n):
    ans = 1
    for i in range(1, n + 1):
        ans = ans * i
    return ans
# Taking input
num = int(input("Enter a positive number: "))
# Call the function and print the result
```

```
print("The factorial is:", factorial(num))
```

Output:

Enter a positive number: 7

The factorial is: 5040

3.Write a user-defined function "find_largest(a, b, c)" that takes three numbers as arguments and returns the largest of the three. Use the function in a program to find and print the largest of three given numbers.

Code:

```
def find_largest(a, b, c):
    return max(a, b, c)


n1 = int(input("Enter first number: "))
n2 = int(input("Enter second number: "))
n3 = int(input("Enter third number: "))
print("Largest number is:", find_largest(n1, n2, n3))
```

Output:

Enter first number: 60

Enter second number: 95

Enter third number: 25

Largest number is: 95

4.Write a user-defined function greet(name) that takes a name as an argument and prints a greeting message like "Hello, [name]!" Use the function to greet the user with their name.

Code:

```
def greet(name):
    print("Hello, " + name + "!")
```

```
user_name = input("Enter your name: ")
greet(user_name)
```

Output:

```
Enter your name: Amman
Hello, Amman!
```

5.Combining Built-in and User-Defined Functions: Find the Average of a List: Write a user-defined function average(numbers) that takes a list of numbers as an argument and returns the average of the numbers. Call the function with a list of numbers and print the average.

Code:

```
def average(numbers):
    return sum(numbers) / len(numbers)


num_str = input("Enter numbers separated by spaces: ").split()
nums = [float(x) for x in num_str]
print("Average is:", average(nums))
```

Output:

```
Enter numbers separated by spaces: 8 4 2
Average is: 4.666666666666667
```

6.Check Palindrome: Write a user-defined function is_palindrome(s) that takes a string as an argument and returns True if the string is a palindrome (reads the same forward and backward), and False otherwise. Test the function with different strings and print the results

Code:

```
def is_palindrome(s):
    return s == s[::-1]
```

```
word = input("Enter a string: ")
print("Is palindrome?", is_palindrome(word))
```

Output:

```
Enter a string: racecar
Is palindrome? True
```

7.Use the Math Module: Write a program that imports the math module and uses it to: Find the square root of a number. Calculate the sine of an angle . Find the greatest common divisor (GCD) of two numbers .

Code:

```
import math
num = float(input("Enter a number for square root: "))
print("Square root:", math.sqrt(num))

angle = float(input("Enter angle in radians: "))
print("Sine:", math.sin(angle))

n1 = int(input("Enter first number for GCD: "))
n2 = int(input("Enter second number for GCD: "))
print("GCD:", math.gcd(n1, n2))
```

Output:

```
Enter a number for square root: 55
Square root: 7.416198487095663
Enter angle in radians: 90
Sine: 0.8939966636005579
Enter first number for GCD: 5
Enter second number for GCD: 4
```

GCD: 1

8.Use the Random Module: Write a program that imports the random module and uses it to: Generate and print a random integer between 1 and 100. Create a list of random numbers and print the list. Shuffle a list of numbers and print the shuffled list.

Code:

```
import random
print("Random integer 1-100:", random.randint(1, 100))

# Create and print list of 5 random numbers
rand_list = []
for i in range(5):
    rand_list.append(random.randint(1, 50))
print("List of random numbers:", rand_list)

# Shuffle list
my_list = [10, 20, 30, 40, 50]
random.shuffle(my_list)
print("Shuffled list:", my_list)
```

Output:

Random integer 1-100: 50

List of random numbers: [48, 49, 50, 16, 31]

Shuffled list: [50, 30, 40, 10, 20]

9.Use the Datetime Module: Write a program that imports the datetime module and uses it to: Get and print the current date and time . Calculate and print the number of days between two dates. Format and print the current date in the format "DD-MM-YYYY"

Code:

```
from datetime import datetime, date
now = datetime.now()
print("Current date and time:", now)

date1 = date(2024, 1, 1)
date2 = date(2024, 12, 31)
diff = date2 - date1
print("Days between dates:", diff.days)

print("Formatted date:", now.strftime("%d-%m-%Y"))
```

Output:

```
Current date and time: 2026-04-26 04:33:39.143792
Days between dates: 365
Formatted date: 26-04-2026
```

10.Use the OS Module: Write a program that imports the os module and uses it to: Print the current working directory Create a new directory and verify its existence. List all files and directories in the current directory .

Code:

```
import os
#print working directory
print(os.getcwd())
#create new directory
os.mkdir('new_folders')
#checking
print(os.path.exists('new_folders'))
#list directory
```

```
print(os.listdir())
import os
#print working directory
print(os.getcwd())
#create new directory
os.mkdir('new_folders')
#checking
print(os.path.exists('new_folders'))
#list directory
print(os.listdir())
```

Output:

True

['arithmetic.py', 'Factorial.py', 'geometry.py', 'main operations.py', 'new_folder', 'new_folders', 'Q10.py', 'Q3.py', 'Q4.py', 'Q5.py', 'Q6.py', 'Q7.py', 'Q8.py', 'Q9.py', 'stringmain.py', 'stringoperations.py', 'stringvalid.py', 'StrlistTup.py', '__pycache__']

11.Create and Use a Custom Package: Create a package called my_math with two modules: arithmetic.py and geometry.py. In arithmetic.py, define functions for addition, subtraction, multiplication, and division. In geometry.py, define functions to calculate the area of a circle and the area of a rectangle. Write a program that imports these functions from the my_math package and uses them to perform various calculations.

Code:

```
#arithmetic.py
def add(a,b):
   return a+b
def subtract(a,b):
   return a- b
def multiply(a,b):
   return a*b
def divide(a, b):
   return a / b
```

```
#geometry.py
def circle_area(radius):
    return 3.14*radius*radius
def rectangle_area(length, width):
    return length * width
#main operations
from operator import add
from arithmetic  import subtract, multiply,divide
from geometry import circle_area,rectangle_area
#arimetic operations
print('Addition:', add(10, 5))
print('Subtraction:', subtract(10,5))
print('Multiplication:', multiply(10, 5))
print('Division:', divide(10, 5))
#Geometry calculations
print('Circle Area:', circle_area(7))
print('Rectangle Area:', rectangle_area(4, 6))
```

Output:

```
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2.0
Circle Area: 153.86
Rectangle Area: 24
```

12.Create a Package for String Utilities: Create a package called string_utils with two modules: string_operations.py and string_validations.py. In string_operations.py, define functions for reversing a string, converting a string to uppercase, and finding the length of a string. In string_validations.py, define functions to check if a string is a palindrome and if it contains only alphabetic characters. Write a program that imports and uses these functions from the string_utils package.

Code:

```
#string operations
def reverse_string(s):
    return s [::-1]
def to_uppercase(s):
    return s.upper()
def string_length(s):
    return len(s)

#stringvalid
def is_palindrome(s):
    return s == s[::-1]
def is_alpha(s):
    return s.isalpha()

#stringsmain
from stringoperations import reverse_string, to_uppercase, string_length
from stringvalid import is_alpha, is_palindrome
from Daily_assignments.DAY3.qn6 import is_palindrome
text = "madam"
print('Original:', text)
print('Reversed:', reverse_string(text))
print('Uppercase:', to_uppercase(text))
print('Length:', string_length(text))
print('Is Palindrome:', is_palindrome(text))
print('Is Alphabet Only:', is_alpha(text))
```

Output:

Enter a string: Amman

Is Palindrome: False

Original: madam

Reversed: madam

Uppercase: MADAM

Length: 5

Is Palindrome: True

Is Alphabet Only: True